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ULTRASOUND SEQUENCING SYSTEM AND METHOD

Abstract

A system comprises a catheter configured for delivery to a body cavity defined by surrounding tissue; a plurality of ultrasound transducers coupled to a distal end of the catheter; and an electronics module configured to selectively turn on/off each ultrasound transducer according to a predetermined activation sequence and to process signals received from each ultrasound transducer to produce at least a 2D display of the surrounding tissue. A user can selectively calculate and display various aspects of cardiac activity. The user can display Dipole Density (DDM), Charge Density (CDM), or Voltage (V-V). The shape and location of the chamber (surface), and the potentials recorded at electrodes can be displayed. The system can also change back and forth between the different display modes, and with post processing tools, can change how various types of information is displayed. Methods are also provided.

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Background/Summary

RELATED APPLICATIONS [0001] The present application is a continuation application of U.S. patent application Ser. No. 17/735,285 filed May 3, 2022, which is a continuation application of U.S. patent application Ser. No. 15/569,185 filed Oct. 25, 2017, which is a 371 national stage application of Patent Cooperation Treaty Application No. PCT/US16/32017 filed May 12, 2016, entitled “Ultrasound Sequencing System And Method”, which in turn claims priority under 35 USC 119 (e) to U.S. Provisional Patent Application Ser. No. 62/160,529, entitled “Ultrasound Sequencing System and Method”, filed May 12, 2015, which is incorporated herein by reference in its entirety. [0002] The present application, while not claiming priority to, may be related to U.S. patent application Ser. No. 14/865,435, entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, filed Sep. 25, 2015, which is a continuation of U.S. Pat. No. 9,167,982 (hereinafter the '982 patent), entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, issued Oct. 27, 2015, which is a continuation of, which is a continuation of U.S. Pat. No. 8,918,158 (hereinafter the '158 patent), entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, issued Dec. 23, 2014, which is a continuation of U.S. Pat. No. 8,700,119 (hereinafter the '119 patent), entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, issued Apr. 15, 2014, which is a continuation of U.S. Pat. No. 8,417,313 (hereinafter the '313 patent), entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, issued Apr. 9, 2013, which was a 35 USC 371 national stage filing of Patent Cooperation Treaty Application No. CH2007/000380, entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls”, filed Aug. 3, 2007, published as WO2008/014629, which claimed priority to Swiss Patent Application No. 1251/06 filed Aug. 3, 2006, each of which is hereby incorporated by reference. [0003] The present application, while not claiming priority to, may be related to U.S. patent application Ser. No. 14/886,449, entitled “Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall”, filed Oct. 19, 2015, which is a continuation of U.S. Pat. No. 9,192,318, entitled “Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall”, issued Nov. 24, 2015, which is a continuation of U.S. Pat. No. 8,512,255, entitled “Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall”, issued Aug. 20, 2013, published as US2010/0298690 (hereinafter the '690 publication), which was a 35 USC 371 national stage application of Patent Cooperation Treaty Application No. PCT/IB09/00071 filed Jan. 16, 2009, entitled “A Device and Method for the Geometric

Determination of Electrical Dipole Densities on the Cardiac Wall”, published as WO2009/090547, which claimed priority to Swiss Patent Application 00068/08 filed Jan. 17, 2008, each of which is hereby incorporated by reference. [0004] The present application, while not claiming priority to, may be related to U.S. patent application Ser. No. 14/003,671, entitled “Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall”, filed Sep. 6, 2013, which is a 35 USC 371 national stage filing of Patent Cooperation Treaty Application No. PCT/US2012/028593, entitled “Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall”, published as WO2012/122517 (hereinafter the '517 publication), which claimed priority to U.S. Patent Provisional Application Ser. No. 61/451,357, each of which is hereby incorporated by reference. [0005] The present application, while not claiming priority to, may be related to U.S. Design application Ser. No. 29/475,273, entitled “Catheter System and Methods of Medical Uses of Same, Including Diagnostic and Treatment Uses for the Heart”, filed Dec. 2, 2013, which is a 35 USC 371 national stage filing of Patent Cooperation Treaty Application No. PCT/US2013/057579, entitled “Catheter System and Methods of Medical Uses of Same, Including Diagnostic and Treatment Uses for the Heart”, filed Aug. 30, 2013, which claims priority to U.S. Patent Provisional Application Ser. No. 61/695,535, entitled “System and Method for Diagnosing and Treating Heart Tissue”, filed Aug. 31, 2012, which is hereby incorporated by reference. [0006] The present application, while not claiming priority to, may be related to U.S. patent application Ser. No. 14/762,944, entitled “Expandable Catheter Assembly with Flexible Printed Circuit Board (PCB) Electrical Pathways”, filed Jul. 23, 2015, which is a 35 USC 371 national stage filing of Patent Cooperation Treaty Application No. PCT/US2014/15261, entitled “Expandable Catheter Assembly with Flexible Printed Circuit Board (PCB) Electrical Pathways”, filed Feb. 7, 2014, published as WO2014/124231, which claims priority to U.S. Patent Provisional Application Ser. No. 61/762,363, entitled “Expandable Catheter Assembly with Flexible Printed Circuit Board (PCB) Electrical Pathways”, filed Feb. 8, 2013, which is hereby incorporated by reference. [0007] The present application, while not claiming priority to, may be related to Patent Cooperation Treaty Application No. PCT/US2015/11312, entitled “Gas-Elimination Patient Access Device”, filed Jan. 14, 2015, which claims priority to U.S. Patent Provisional Application Ser. No. 61/928,704, entitled “Gas-Elimination Patient Access Device”, filed Jan. 17, 2014, which is hereby incorporated by reference. [0008] The present application, while not claiming priority to, may be related to Patent Cooperation Treaty Application No. PCT/US2015/22187, entitled “Cardiac Analysis User Interface System and Method”, filed Mar. 24, 2015, which claims priority to U.S. Patent Provisional Application Ser. No. 61/970,027, entitled “Cardiac Analysis User Interface System and Method”, filed Mar. 28, 2014, which is hereby incorporated by reference. [0009] The present application, while not claiming priority to, may be related to U.S. application Ser. No. 14/916,056, entitled “Devices and Methods for Determination of Electrical Dipole Densities on a Cardiac Surface”, filed Mar. 2, 2016, which is a 35 USC 371 national stage filing of Patent Cooperation Treaty Application No. PCT/US2014/54942, entitled “Devices and Methods for Determination of Electrical Dipole Densities on a Cardiac Surface”, filed Sep. 10, 2014, published as WO2015/038607, which claims priority to U.S. Patent Provisional Application Ser. No. 61/877,617, entitled “Devices and Methods for Determination of Electrical Dipole Densities on a Cardiac Surface”, filed Sep. 13, 2013, which is hereby incorporated by reference.

FIELD

[0010] The present invention is generally related to systems and methods that may be useful for the diagnosis and/or treatment of cardiac arrhythmias or other cardiac diseases or disorders, such as systems, devices, and methods that may be useful in mapping cardiac activity.

BACKGROUND

[0011] For localizing the origin(s) of cardiac arrhythmias it is common practice to measure the

electric potentials located on the inner surface of the heart by electrophysiological means within the patient's heart. One method is to insert electrode catheters into the heart to record cardiac potentials during normal heart rhythm or cardiac arrhythmia. If the arrhythmia has a regular activation sequence, the timing of the electric activation measured in voltage at the site of the electrode can be accumulated when moving the electrode around during the arrhythmia, to create a three-dimensional map of the electric activation. By doing this, information on the location of the source of arrhythmia(s) and mechanisms, i.e., re-entrant circuits, can be diagnosed to initiate or guide treatment (radiofrequency ablation). The information can also be used to guide the treatment of cardiac resynchronization, in which implantable pacing electrodes are placed in specific locations within the heart wall or chambers to re-establish a normal level of coordinated activation of the heart.

[0012] A method using external sensors measures the electrical activity of the heart from the body surface using electrocardiogram techniques that include, for example, electrocardiograms (ECG) and vectorcardiography (VCG). These external sensor techniques can be limited in their ability to provide information and/or data on regional electrocardiac activity. These methods can also fail to localize bioelectric events in the heart.

[0013] A method using external sensors for the localization of cardiac arrhythmias utilizes body surface mapping. In this technique, multiple electrodes are attached to the entire surface of the thorax and the information of the cardiac electrograms (surface ECG) is measured in voltages that are accumulated into maps of cardiac activation. This measurement can be problematic because the electrical activity is time dependent and spatially distributed throughout the myocardium and also fails to localize bioelectric events in the heart. Complex mathematical methods are required to determine the electrical activation upon the outer surface of a heart model (i.e. epicardium), for instance, one obtained from CT or MRI imaging giving information on cardiac size and orientation within the thoracic cavity.

[0014] Alternatively, recordings of potentials at locations on the torso, for example, can provide body surface potential maps (BSPMs) over the torso surface. Although the BSPMs can indicate regional cardiac electrical activity in a manner that can be different from conventional ECG techniques, these BSPM techniques generally provide a comparatively low resolution, smoothed projection of cardiac electrical activity that does not facilitate visual detection or identification of cardiac event locations (e.g., sites of initiation of cardiac arrhythmias) and details of regional activity (e.g., number and location of arrhythmogenic foci in the heart).

[0015] Since the localization of cardiac arrhythmias by the use of potentials is imprecise, the successful treatment of cardiac arrhythmias has been difficult and has demonstrated limited success and reliability. There is, therefore, a need for improved methods of localizing, diagnosing and treating cardiac arrhythmias.

SUMMARY

[0016] In accordance with one aspect of the inventive concept, provided is a body cavity imaging system, comprising: a catheter configured for delivery to a body cavity defined by surrounding tissue; a plurality of ultrasound transducers coupled to a distal end of the catheter; an electronics module configured to selectively turn on/off each ultrasound transducer according to a predetermined activation sequence and to process signals received from each ultrasound transducer to produce at least a 2D display of the surrounding tissue.

[0017] In various embodiments, the imaging system can be part of an electrophysiology system.

[0018] In various embodiments, the cavity can be a heart chamber and the surrounding tissue can be one or more walls of the heart chamber.

[0019] In various embodiments, the display can be a 3D display of the surrounding tissue.

[0020] In various embodiments, the 3D display of the surrounding tissue can be presented on a user interface system having a display screen and user control mechanism enabling graphical manipulation of the 3D display of the surrounding tissue.

[0021] In various embodiments, the graphical manipulation can include one or more of zoom in/out, rotate, select portions or subsections of the surrounding tissue.

[0022] In various embodiments, the plurality of ultrasound transducers can be coupled to a 3D array.

[0023] In various embodiments, the 3D array can be a basket array, spiral array, a balloon, radially deployable arms, and/or other expandable and compactible structures.

[0024] In various embodiments, the ultrasound transducers can be disposed on a plurality of splines of the 3D array.

[0025] In various embodiments, the 3D array can include at least three splines.

[0026] In various embodiments, at least two ultrasound transducers can be disposed on each spline.

[0027] In various embodiments, the system can further comprise a plurality of biopotential electrodes coupled to a distal end of the catheter.

[0028] In various embodiments, the biopotential electrodes can also be disposed on a plurality of splines of the 3D array.

[0029] In various embodiments, at least some of the biopotential electrodes and at least some of the ultrasound transducers can be disposed on the same splines.

[0030] In various embodiments, a biopotential electrode and an ultrasound transducer are disposed together to form an electrode/transducer pair, and the system includes a plurality of electrode/transducer pairs.

[0031] In various embodiments, one or more splines can comprise at least one electrode/transducer pair.

[0032] In various embodiments, one or more splines can comprise a plurality of electrode/transducer pairs.

[0033] In various embodiments, a plurality of splines can comprise at least one electrode/transducer pair.

[0034] In various embodiments, a plurality of splines can comprise a plurality of electrode/transducer pairs.

[0035] In various embodiments, a plurality of splines can comprise at least three electrode/transducer pairs.

[0036] In various embodiments, each spline can comprise a flexible PCB, and each electrode/transducer pair is electrically coupled to the flexible PCB.

[0037] In various embodiments, each electrode/transducer pair can share a common communication path on the flexible PCB.

[0038] In various embodiments, all electrode/transducer pairs on a spline can share a common communication path on the flexible PCB.

[0039] In various embodiments, the common communication path can be a common ground.

[0040] In various embodiments, the system can be further configured to correlate cardiac or other electrical activity to one or more images generated using imaging device.

[0041] In various embodiments, the imaging device can comprise an imaging device selected from the group consisting of: a fluoroscope; an MRI; a CT Scanner; an ultrasound imaging device; and combinations of two or more of these.

[0042] In various embodiments, the activation sequence can be a pattern of turning on/off the plurality of ultrasound transducers that avoids the sequential activation of two neighboring ultrasound transducers.

[0043] In various embodiments, the activation sequence can avoid the sequential activation of two transducers within two or three neighboring spaces of each other.

[0044] In various embodiments, the neighboring spaces can be considered spaces on a single spline; across splines, such as transducer 1 of spline 1 and transducer 1 of spline 2; and/or diagonally across splines, such as transducer 1 of spline 1 and transducer 2 of spline 2.

[0045] In various embodiments, the activation sequence pattern can be a pattern that avoids

sequential activation of two transducers from a single spline.

[0046] In accordance with another aspect of the inventive concept, provided is a method of performing a diagnostic assessment, comprising: providing a cardiac diagnostic system, including a plurality of ultrasound transducers and a plurality of electrodes coupled to the end of a diagnostic catheter; inserting the diagnostic catheter into a heart chamber of a patient; placing the cardiac diagnostic system in a diagnostic mode; performing a biopotential measurement process; performing a localization process; performing an ultrasound measurement process; and interleaving a localization process and the ultrasound process.

[0047] In various embodiments, frequencies of the ultrasound transducers do not interfere with biopotential signals and biopotential signals do not interfere with localization signals.

[0048] In various embodiments, the biopotential measurement process can be performed continuously.

[0049] In various embodiments, the biopotential measurement process can be interleaved with the localization process and the ultrasound measurement process.

[0050] In various embodiments, the method can comprise performing the localization process longer than, or multiple times for, a single ultrasound measurement process.

[0051] In various embodiments, the method can comprise performing the ultrasound measurement process longer than, or multiple times for, a single localization process.

[0052] In various embodiments, the biopotential measurement process can include measuring and analyzing biopotentials from the electrodes.

[0053] In various embodiments, the biopotential measurement process can include determining dipole densities and/or surface charge densities from the biopotential data.

[0054] In accordance with another aspect of the inventive concepts, provided is a method of performing a localization process, comprising: providing a cardiac diagnostic system, including a plurality of biopotential electrodes and, optionally, a plurality of ultrasound transducers coupled to a distal end of a catheter; inserting the diagnostic catheter into a heart chamber of a patient; placing one or more pairs of surface electrodes on the patient and defining an individual axis for each pair of electrodes; generating one or more localization signals and transmitting same to the patient through the one or more pairs of surface electrodes; recording data collected from the one or more pairs of surface electrodes; filtering the recorded data to isolate signals correlating to the generated localization signals of each pair of surface electrodes; analyzing the filtered data to determine a location of each biopotential electrode in a coordinate system relative to the patient, the coordinate system defined by the one or more pairs of surface electrodes.

[0055] In various embodiments, there can be at least two pairs of electrodes, and one individual axis can be determined for each pair of surface electrodes.

[0056] In various embodiments, there can be at least three pairs of electrodes, and one individual axis is determined for each pair of surface electrodes.

[0057] In various embodiments, the three axes can define a three axis localization system.

[0058] In various embodiments, the coordinate system can be a 3D coordinate system.

[0059] In various embodiments, an origin of the coordinate system can be logically located within the heart of the patient.

[0060] In various embodiments, the method can comprise: placing surface electrodes from a first pair on the chest and back of the patient, defining a first axis; and/or placing surface electrodes from a second pair laterally on the sides of the patient, defining a second axis; and/or placing surface electrodes from a third pair on the neck or shoulder and thigh of the patient, defining a third axis.

[0061] In various embodiments, the method can comprise: placing surface electrodes from a first pair of electrodes laterally on the sides of the patient, defining a first axis; and/or placing surface electrodes from a second pair of electrodes on the upper chest and lower back of the patient, defining a second axis; and/or placing surface electrodes from a third pair of electrodes on the

upper back and lower chest of the patient, defining a third axis.

[0062] In various embodiments, each pair of surface electrodes can be individually driven with a signal having a different frequency.

[0063] In various embodiments, localization signals can be generated at a frequency in a range of about 1-100 KHz.

[0064] In various embodiments, the signals from each pair of surface electrodes can be individually recorded.

[0065] In various embodiments, the signals from each pair of surface electrodes can be individually filtered.

[0066] In various embodiments, the localization process can be interleaved with an ultrasound measurement process of the cardiac diagnostic system.

[0067] In various embodiments, the localization process can be interleaved with a biopotential measurement process of the cardiac diagnostic system.

[0068] In accordance with aspects of the inventive concept, provided is a method of performing an ultrasound measurement process, comprising: providing a cardiac diagnostic system, including a plurality of ultrasound transducers and, optionally, a plurality of biopotential electrodes coupled to a distal end of a catheter; inserting the diagnostic catheter into a heart chamber; activating (or ringing) an ultrasound transducer to generate an ultrasound transducer signal; ringing down the ultrasound transducer; sensing and recording a reflection of the ultrasound transducer signal by a source; determining a distance from the transducer to the source based on the received reflection; repeating the above steps until all ultrasound transducers have been activated; and repeating the above steps for all ultrasound transducers until the ultrasound measurement process is complete or ended.

[0069] In various embodiments, the biopotential electrodes and ultrasound transducers can be paired to form electrode/transducer pairs.

[0070] In various embodiments, the electrode/transducer pairs can be disposed on a plurality of splines of a 3D array.

[0071] In various embodiments, activating an ultrasound transducer can include closing one or more switches, thereby electrically connecting the transducer to a signal generator.

[0072] In various embodiments, the one or more switches can comprise an opto-coupler.

[0073] In various embodiments, the opto-coupler can have an activation time in a range of about 0.01 μ s, or 500 μ s.

[0074] In various embodiments, activating the transducer can include

[0075] generating a pulsed drive signal configured to ring, vibrate, and/or otherwise cause the transducer to generate an ultrasonic pulse.

[0076] In various embodiments, the drive signal can comprise a signal with a frequency in a range of about 1 MHz and 25 MHz, such as 10 MHz.

[0077] In various embodiments, the drive signal frequency can be about 10 MHz.

[0078] In various embodiments, the drive signal can further comprise a signal with a pulse width in a range of about 0.1 μ s and 10 μ s.

[0079] In various embodiments, the drive signal pulse width can be about 2.0 μ s.

[0080] In various embodiments, the ring down can have a duration of between about 0.05 μ s and 1 μ s for dissipation of vibration of the ultrasound transducer.

[0081] In various embodiments, the ring down can have a duration of about 0.1 μ s.

[0082] In various embodiments, sensing the reflection can be performed for a duration in a range of about 1 μ s and 200 μ s.

[0083] In various embodiments, the sensing duration can be about 100 μ s.

[0084] In various embodiments, the source can be an inner wall of a cardiac chamber.

[0085] In various embodiments, the activation of the transducer can cause deactivation of a paired biopotential electrode.

[0086] In various embodiments, the method can further comprise non-sequentially activating electrode/transducer pairs, thereby not broadening a temporary “blind spot” of a neighboring biopotential electrode caused by the activation of the ultrasound transducer.

[0087] In various embodiments, the patient can be a living being.

[0088] In various embodiments, the patient can be a simulated being or heart.

[0089] In accordance with aspects of the inventive concept, provided is a body cavity imaging system as shown and/or described.

[0090] In accordance with aspects of the inventive concept, provided is a cardiac diagnostic system as shown and/or described.

[0091] In accordance with aspects of the inventive concept, provided is a cardiac diagnostic process as shown and/or described.

[0092] In accordance with aspects of the inventive concept, provided is a localization process as shown and/or described.

[0093] In accordance with aspects of the inventive concept, provided is a biopotential measurement process as shown and/or described.

[0094] In accordance with aspects of the inventive concept, provided is an ultrasound imaging method as shown and/or described.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0095] FIG. 1 illustrates a schematic view of an exemplary embodiment of a cardiac analysis system comprising a catheter with an assembly including multiple electrical components that can be deployed within a body, in accordance with aspects of the present inventive concepts.

[0096] FIG. 2 provides a flowchart of an embodiment of a method of performing a diagnostic assessment, in accordance with aspects of the present inventive concepts.

[0097] FIG. 3 provides a flowchart of an embodiment of a method of performing a localization process, in accordance with aspects of the present inventive concepts.

[0098] FIG. 4 provides a flowchart of an embodiment of a method of performing an ultrasound measurement process, in accordance with aspects of the present inventive concepts.

[0099] FIG. 5 provides a perspective view of an embodiment of a diagnostic catheter, in accordance with aspects of the present inventive concepts.

[0100] FIG. 5A is a perspective view of the catheter of FIG. 5 in an altered shape, in accordance with aspects of the present inventive concepts.

[0101] FIG. 6 provides a representation of an embodiment of an activation sequence of an array of ultrasound transducers disposed on six splines, in accordance with aspects of the present inventive concepts.

[0102] FIG. 7 provides an embodiment of a block diagram of a user interface system that can be used with a diagnostic catheter as described herein, for example, in accordance with the present inventive concepts.

[0103] FIGS. 8A-8C provide different views relating to the output of the user interface system, in accordance with aspects of the present inventive concepts.

[0104] FIG. 9 provides a functional block diagram of an embodiment of a cardiac information processing system, in accordance with the present inventive concepts.

DETAILED DESCRIPTION

[0105] Various exemplary embodiments will be described more fully hereinafter with reference to the accompanying drawings, in which some exemplary embodiments are shown. The present inventive concepts can, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein.

[0106] It will be understood that, although the terms first, second, etc. are used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another, but not to imply a required sequence of elements. For example, a first element can be termed a second element, and, similarly, a second element can be termed a first element, without departing from the scope of the present invention. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. And a “combination” of associated listed items need not include all of the items listed, but can include all of the items listed.

[0107] It will be understood that when an element is referred to as being “on” or “attached”, “connected” or “coupled” to another element, it can be directly on or connected or coupled to the other element or intervening elements can be present. In contrast, when an element is referred to as being “directly on” or “directly connected” or “directly coupled” to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.).

[0108] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used herein, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0109] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like can be used to describe an element and/or feature's relationship to another element(s) and/or feature(s) as, for example, illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use and/or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” and/or “beneath” other elements or features would then be oriented “above” the other elements or features. The device can be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0110] Various exemplary embodiments are described herein with reference illustrations of idealized or representative structures and intermediate structures. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, exemplary embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0111] To the extent that functional features, operations, and/or steps are described herein, or otherwise understood to be included within various embodiments of the present inventive concepts, such functional features, operations, and/or steps can be embodied in functional blocks, units, modules, operations and/or methods. And to the extent that such functional blocks, units, modules, operations and/or methods include computer program code, such computer program code can be stored in a computer readable medium, e.g., such as non-transitory memory and media, that is executable by at least one computer processor.

[0112] Referring now to FIG. 1, a schematic view of an embodiment of a cardiac analysis system comprising a catheter with an assembly including multiple electrical components that can be deployed within a body is illustrated, consistent with the present inventive concepts. System 10 includes diagnostic catheter 100 and electronics module 200. In some embodiments, system 10 can further include an introducer 50 and/or imaging device 80. Introducer 50 comprises handle 51 and elongate shaft 55. Shaft 55 comprises at least one lumen, such as a lumen configured to slidably

receive diagnostic catheter **100** within shaft **55**. In some embodiments introducer **50** comprises a transeptal access sheath or other device configured to provide access to a body space or cavity, such as a heart chamber, for example. Handle **51** can include a knob, lever, switch or other control, generally referred to herein as control **52**. Control **52** can be configured to steer or otherwise deflect the distal end of introducer **50**. Imaging device **80** can comprise an imaging device selected from the group consisting of: a fluoroscope; an MRI; a CT Scanner; an ultrasound imaging device; and combinations of two or more of these. However, other imaging devices could be used in various embodiments.

[0113] Diagnostic catheter **100** includes handle **110**, and an elongate flexible shaft, shaft **105**, extending from handle **110**. Attached to the distal end of shaft **105** is a radially expandable and/or compactable assembly, expandable assembly **130**. In an alternative embodiment, expandable assembly **130** is mounted to (e.g. surrounding) a distal portion of shaft **105**, at a location proximal to the distal end of shaft **105**. In some embodiments, expandable assembly **130** is constructed and arranged as described in reference to applicant's co-pending U.S. patent application Ser. No. 14/422,941, titled "System and Method for Diagnosing and Treating Heart Tissue", filed Feb. 5, 2015, the content of which is incorporated herein by reference in its entirety. Shaft **105** and expandable assembly **130** are constructed and arranged to be inserted into a body (e.g. an animal body or a human body, such as the body of Patient P), and advanced through a body vessel, such as a femoral vein, jugular vein, or other blood vessel. Shaft **105** and expandable assembly **130** can be constructed and arranged to be inserted through introducer **50**, such as when expandable assembly **130** is in a compacted state, and slidingly advanced through a lumen of shaft **55** into a body space, such as a chamber of the heart, such as the right atrium or the left atrium, as examples.

[0114] Handle **110** can include one or more controls, such as control **111**. Control **111** can comprise a knob, switch, lever, button, slide, or other control configured to perform a function selected from the group consisting of: steer the distal portion of shaft **105**; control the expansion and/or contraction of expandable assembly **130** such as by advancing and/or retracting a control rod, not shown but such as is described herebelow in reference to FIG. 5; control the shape of expandable assembly **130**, such as by advancing or retracting a control rod operably attached to expandable assembly **130**; close and/or open an electrical connection, such as to provide power to one or more components of expandable assembly **130**; initiate a process or otherwise send a command or other user activated signal to electronics module **200**; and combinations of these.

[0115] Expandable assembly **130** can comprise a structure including multiple flexible arms or splines, splines **131a-c** (singly or collectively splines **131**), as shown. In some embodiments, expandable assembly **130** can comprise between two and ten splines **131**, such as six splines **131**. In the embodiment of FIG. 1, three splines **131a-c** are equally spaced about a central axis of catheter **100** (i.e., a spacing of 120° between each spline when expandable assembly **130** is deployed in its expanded state). In other embodiments, splines **131** can be equally or unequally spaced, such as two, four, eight or twelve splines **131** with an equal spacing of 180°, 90°, 60°, 45°, and/or 30°, respectively. In some embodiments, expandable assembly **130** can comprise a balloon, radially deployable arms, and/or other expandable and compactible structure.

[0116] Expandable assembly **130** can further comprise multiple "pairs" of electrical components, for example, at least one pair comprising an electrode **132** and an ultrasound element, transducer **133**. Each electrode **132** can be configured to record a voltage, such as the voltage present on a surface of the heart or at a location within a heart chamber. Each ultrasound transducer **133** can be configured to send and/or receive ultrasound signals, such as to produce an anatomical image of the tissue of at least a portion of the heart or other patient anatomical location. Electrodes **132** and ultrasound transducers **133** can comprise different shapes, such as a shape selected from the group consisting of: round; triangular; rectangular; hexagonal; trapezoidal; and combinations of two or more of these. In some embodiments, a first electrode **132** has a different shape than a second electrode **132**. In some embodiments, a first ultrasound transducer **133** has a different shape than a

second ultrasound transducer **133**. In some embodiments, one or more ultrasound transducers **133** each comprise a single element or an array of elements (e.g. a microarray of ultrasound elements), for example an array of ultrasound elements configured as a phased array (e.g. to allow steering and/or focusing of ultrasound energy). In some embodiments, one or more ultrasound transducers **133** comprise an element selected from the group consisting of: bulk ceramic (thickness-mode or spherical); micromachined ultrasound transducer (MUT), such as piezoelectric (pMUT) or capacitive (cMUT); thin film such as PVDF; shear-wave; and combinations of two or more of these.

[0117] Each connected pair of an electrode **132** and an ultrasound transducer **133** can share a single conductor (e.g. a wire or other communication and/or power delivery conduit), such as communication path **134** (e.g. a wire) described herebelow. In some embodiments, multiple pairs of electrode **132** and ultrasound transducer **133** can collectively share a single conductor, communication path **135** (e.g. a wire), also as described herebelow.

[0118] The embodiment of FIG. **1** shows three electrode/transducer pairs per each spline **131a-c** (i.e. nine pairs for expandable assembly **130**), each pair comprising an electrode **132** and an ultrasound transducer **133**. Spline **131a** comprises three electrode/ultrasound pairs, **132.sub.i/133.sub.i-132.sub.iii/133.sub.iii**. Spline **131b** comprises three electrode/ultrasound pairs, **132.sub.iv/133.sub.iv-132.sub.vi/133.sub.vi**. Spline **131c** comprises three electrode/ultrasound pairs, **132.sub.vii/133.sub.vii-132.sub.ix/133.sub.ix**. Each electrode/ultrasound pair **132/133** is electrically or otherwise operably connected to a connection point **136** via a communication path **134**, such as when splines **131** include a printed circuit (e.g. a flexible printed circuit), and communication paths **134** can comprise traces on the printed circuit, such as is described in reference to applicant's co-pending U.S. patent application Ser. No. 14/762,944, titled "Expandable Catheter Assembly with Flexible Printed Circuit Board (PCB) Electrical Pathways", filed Jul. 23, 2015, the content of which is incorporated herein by reference in its entirety. In various embodiments, such as the embodiment shown in FIG. **1**, one or more electrode/ultrasound pairs **132/133** can share a common communication path **135**, such as a trace configured as a common ground, electrically or otherwise operably connected to a connection point **136**.

[0119] In the embodiment shown, a communication path **134** is connected to an electrode **132**, such as electrode **132.sub.i**, which is connected to the positive terminal of a paired ultrasound transducer **133**, such as ultrasound transducer **133.sub.i**. The negative terminal of ultrasound transducer **133.sub.i** is connected to common communication path **135**. In some embodiments, two or more electrode/ultrasound pairs **132/133** can share a common communication path **135**. In some embodiments, each spline **131** can comprise two or more common communication paths **135**, such as a spline comprising eight electrode/ultrasound pairs **132/133**, comprising two common communication paths **134**, each shared by four electrode/ultrasound pairs **132/133**.

[0120] A conduit comprising one or more electrical, optical, or electro-optical wires or cables (e.g. coaxial wires), such as conduit **106**, can provide a communication path between one or more components of expandable assembly **130**, such as one or more electrode/ultrasound pairs **132/133**, and handle **110** of catheter **100**. Conduit **106** terminates in handle **110** at connector **116**. Connector **116** can comprise a jack, plug, terminal, port, or other custom or standard electrical, optical, or electro-optical connector. Conduit **106** can extend distally from handle **110**, through one or more lumens of shaft **105**, and terminate at the one or more connection points **136**. In some embodiments, conduit **106** can comprise multiple coaxial cables, configured to extend through multiple lumens within shaft **105**, such as when conduit **106** comprises one coaxial cable per electrode/ultrasound pair **132/133**, and the coaxial shields are constructed and arranged to provide a common communication medium (e.g. a ground wire). Two or more coaxial cables can be joined to share a common communication medium, such as four or eight coaxial cables linked to create a common channel. In some embodiments, a coaxial cable can be used that comprises a gauge greater than 36AWG, such as 42AWG or 46AWG, and can comprise a nominal impedance of less

than or equal to 500, and a capacitance of approximately 110 pF/m at 1 KHz.

[0121] Electronics module **200** comprises one or more connectors **216**, each comprising a jack, plug, terminal, port, or other custom or standard electrical, optical, or electro-optical connector. System **10** can comprise a cable or other conduit, such as cable **206**, configured to electrically, optically, and/or electro-optically connect catheter **100** to electronics module **200** via connectors **116** and **216**. In some embodiments, electronics module **200** can comprise a patient isolation circuit **201**, configured to electrically isolate one or more components of electronics module from Patient P (e.g. to prevent undesired delivery of a shock or other undesired electrical energy to Patient P). Isolation circuit **201** can be integral to electronics module **200** and/or it can comprise a separate discrete component (e.g. separate housing).

[0122] System **10** can further comprise one or more surface electrodes **225**, e.g., such as patch electrodes configured to attach to the skin of the patient. Surface electrodes **225** are electrically connected to electronics module **200** via one or more electrical, optical or other conduits, referred to as conduits **226**. Surface electrodes can be constructed and arranged to transmit and/or record signals to and/or from Patient P, such as when surface electrodes **225** transmit electrical signals to generate one or more electrical fields within Patient P, such as electrical fields used in a localization procedure as described herein. In some embodiments, system **10** can be configured to generate one or more images based upon information recorded using diagnostic catheter **100**, and to correlate cardiac or other electrical activity (e.g. voltage information, dipole information and/or surface charge information) to the one or more images. Alternatively or additionally, system **10** can be configured to correlate cardiac or other electrical activity to one or more images generated using imaging device **80**.

[0123] Electronics module **200** comprises electrode transceiver circuitry **210**, ultrasound transceiver circuitry **220**, and user interface subsystem **230**. Electrode transceiver circuitry (ETC) **210** comprises one or more components selected from the group consisting of: a processor, such as a computer processor configured to perform one or more calculations based on data recorded from electrodes **132**; at least one filter, such as one or more filters configured to filter one or more data sets recorded from electrodes **132**; at least one signal generator, such as signal generator **211**, configured to generate signals used to create a localization field as described herebelow; at least one memory module, such as a memory module configured to store data recorded from electrodes **132**; and combinations of these.

[0124] Ultrasound transceiver circuitry (UTC) **220** comprises one or more components selected from the group consisting of: a processor, such as a computer processor configured to perform one or more calculations based on data recorded from ultrasound transducers **133**; at least one filter, such as one or more filters configured to filter one or more data sets recorded from transducers **133**; at least one signal generator, such as signal generator **221**, configured to generate signals used to drive transducers **133** to cause an ultrasonic signal to be produced as described herebelow; at least one memory module, such as a memory module configured to store data recorded from transducers **133**; and combinations of these. However, in some embodiments, the ETC **210** and UTC **220** can share components, such as sharing one or more processors and/or one or more memory module.

[0125] User interface subsystem **230** can comprise one or more user input and/or user output components, such as one or more components selected from the group consisting of: a keyboard; a mouse; one or more buttons or switches; a monitor; a touch screen; a speaker; a microphone; a foot pedal; a printer; a transmitter, a receiver, and combinations of these. User interface subsystem **230** can be configured to allow user input, such as to set one or more parameters associated with the operation of system **10**. User interface subsystem can be further configured to display information to a user, such as information selected from the group consisting of: electrical cardiac activity information (e.g., dipole density, surface charge density, and/or voltage information, such as, voltage information measured and recorded from electrodes **132** and/or dipole or surface charge density information calculated from data recorded from electrodes **132**); device localization

(position) data, such as data calculated from data recorded from electrodes **132** and/or other electrodes of system **10**; cardiac geometry data, such as geometry data calculated from signals provided by ultrasound transducers **133**; one or more images, such as one or more images recorded from imaging device **80** and/or one or more images generated by electronics module **200** (e.g. from data provided by ultrasound transducers **133**), such as a text or graphical representation of one or more calculated values by ETC **210** and/or UTC **220**; and combinations of these.

[0126] In some embodiments, system **10** can comprise a system constructed and arranged to determine a dipole density map correlating to the distribution of dipole densities on the wall of a heart chamber, and/or a surface charge density map correlating to the distribution of surface charge densities on the wall of a heart chamber such as the system described in applicant's U.S. Pat. No. 8,512,255, titled "Device and Method for the Geometric Determination of Electrical Dipole Densities on the Cardiac Wall", filed Aug. 31, 2012, the content of which is incorporated herein by reference in its entirety. Alternatively or additionally, system **10** can comprise a system constructed and arranged to determine a voltage map, or other diagnostic data set of electrical or anatomic information recorded by catheter **100** and/or calculated by electronics module **200**.

[0127] Electrodes **132** can be configured to record electrical activity of the heart chamber, such as by biopotentials (voltages) representing the electrical activity of the heart. Electrodes **132** can be further configured to perform a localization process, comprising recording a voltage caused by an electrical field, such as a localization field generated by surface electrodes **225**. Electronics module **200** and ultrasound transducers **133** can be configured to perform an ultrasonically-based distance measurement, comprising transmitting ultrasonic signals from one or more ultrasound transducers **133**, and having similar or dissimilar ultrasound transducers **133** record at least the first reflections of the transmitted signals.

[0128] ETC **210** can be configured to process data recorded by electrodes **132** to produce information selected from the group consisting of: the location of individual electrodes **132**; the location, current geometry and/or orientation of expandable assembly **130** and its respective components (by processing recorded localization data); the location of one or more additional components or devices present within the heart chamber; electrical activity of a heart chamber, such as dipole density or surface charge density on the wall of the heart chamber or voltages, by processing recorded biopotential data; and combinations of these.

[0129] UTC **220** can be configured to process recorded ultrasound reflection data from ultrasound transducers **133** to produce information selected from the group consisting of: distance from a transducer **133** to a first surface of a heart chamber; distance from an ultrasound transducer **133** to a second surface of a heart chamber; distance between a first surface of a heart chamber and a second surface of a heart chamber (e.g. a heart wall thickness comprising the distance between the endocardial surface and epicardial surface of a heart chamber); location of one or more anatomic features, such as the pulmonary veins (e.g. pulmonary vein ostia); location of a cardiac valve; other anatomic geometry information; tissue velocity; tissue density; distance from a transducer **133** to a surface of another component of system **10**; and combinations of these.

[0130] In some embodiments, a single component (e.g. only a single electrode **132** or a single ultrasound transducer **133**) of an electrode **132**/ultrasound transducer **133** pair is "activated" at a time (e.g., is provided a signal by electronics module **200** or has its signal recorded by electronics module **200**). For example, during the activation period of an ultrasound transducer **133** (e.g. comprising ringing, ringing down, and/or recording), the recording and/or driving of its paired electrode **132** can be disabled (e.g. not performed or ignored). Alternatively, during the activation of an electrode **132** (e.g. driving and/or recording), driving or recording of a paired ultrasound transducer **133** can be disabled (e.g., not performed or ignored). Isolation or activation of either an electrode **132** or an ultrasound transducer **133** of a connected pair can prevent issues that can be caused by an ultrasound transducer **133** drive signal interfering with a localization drive signal (e.g. provided by a surface electrode) and/or a biopotential signal recorded by an electrode **132**. In some

embodiments, one or more recorded signals are filtered, allowing for simultaneous operation of ultrasound processing and biopotential processing. In some embodiments, system **10** can comprise a standard diagnostic mode, comprising performing biopotential measurements continuously, and interleaving a localization process and an ultrasound measurement process, such as process **500** described in reference to FIG. **2** herebelow. Ultrasound signals can interfere with biopotential signals, and/or biopotential signals can interfere with localization signals. In some embodiments, one or more processes (localization, ultrasound, and biopotential measurements) can be interleaved with one or more other processes, such that an individual process (or combination of processes) does not cause interference with a separate process (or combination of processes).

[0131] During an operational mode, such as a diagnostic mode as described in FIG. **2** herebelow, the activation period of a transducer **133** causes a “blanked” period for paired electrode **132**, causing a temporary “blind spot” of biopotential measurement. As described in reference to FIG. **4** herebelow, a sequencing of transducers **133** can be performed, such that the temporary “blind spot” is not extended by sequentially activating adjacent or otherwise proximate pairs **132/133**.

[0132] In some embodiments, a sequence is performed as follows. During an ultrasound measurement process, all electrodes **132** can actively record biopotential signals. A first transducer **133_i** can be activated, as described herebelow in reference to FIG. **4**, causing a “blanking” of paired electrode **132.sub.i**. Following the activation of transducer **133.sub.i**, transducer **133.sub.v** can be activated, followed by **133.sub.ix**, **133.sub.ii**, **133.sub.vi**, **133.sub.viii**, **133.sub.iii**, **133.sub.iv**, and **133.sub.vii**. In this embodiment, the “blind spot” created by the “blanking” of paired electrodes **132** follows the same pattern, moving non-sequentially about expandable assembly **130**, and minimizing any potential data integrity loss due to the blind spots created.

[0133] In some embodiments, system **10** comprises one or more sensors, each configured to produce a signal, such as sensor **59** of introducer **50**, a sensor of diagnostic catheter **100** (e.g. sensor **119** of handle **110** or sensor **139** of array **130**), a sensor **209** of electronics module **200** and/or a sensor **89** of imaging device **80**, each as shown in FIG. **1**. In some embodiments, system **10** comprises two or more of sensors **59**, **119**, **139**, **209** and/or **89**. In some embodiments, sensors **59**, **119**, **139**, **209** and/or **89** comprise a sensor selected from the group consisting of: a force sensor; a pressure sensor; a strain gauge; an optical sensor; an imaging sensor (e.g. a lens or optical fiber); a sound sensor such as an ultrasound sensor; a hall effect sensor; a pH sensor; a magnetic sensor; a temperature sensor; and combinations of one or more of these. In some embodiments, sensors **59** and/or **139** comprise a patient physiologic sensor, such as a sensor selected from the group consisting of: a blood pressure sensor; a blood gas sensor; a temperature sensor; a blood glucose sensor; a pH sensor; a respiration sensor; an average clotting time (ACT) sensor; and combinations of one or more of these. In some embodiments, system **10** is configured to analyze a signal produced by one, two or more of sensors **59**, **119**, **139**, **209** and/or **89**. In some embodiments, system **10** (e.g. electronics module **200** and/or an algorithm of ETC **210**) is configured to perform an analysis of one or more signals produced by one, two or more of sensors **59**, **119**, **139**, **209** and/or **89** in combination with voltage data, dipole density data, surface charge data, and/or anatomical data (e.g. anatomical data collected by one or more ultrasound transducers **133**). In some embodiments, signals from one or more sensors **59**, **119**, **139**, **209** and/or **89** are used by system **10** to perform a function selected from the group consisting of: improve an anatomical image displayed by system **10**; improve cardiac information displayed by system **10** (e.g. dipole density and/or surface charge information); detect a malfunction of system **10**; provide physiologic data of a patient; and combinations of one or more of these. In some embodiments, one or more of sensors **59**, **119**, **139**, **209** and/or **89** can comprise a transducer (e.g. as an alternative to being a sensor or in addition to being a sensor), such as a transducer selected from the group consisting of: a heating element; a cooling element; a vibrating element; a drug or other agent delivery element; a magnetic field generating element; a light delivery element; an imaging element (such as a lens, and/or optical fiber); and combinations of one or more of these.

[0134] Referring now to FIG. 2, provided is an embodiment of a method of performing a diagnostic assessment, consistent with the present inventive concepts. In some embodiments, process **500** of FIG. 2 is accomplished using system **10** of FIG. 1 described hereabove. In STEP **510**, a diagnostic catheter **100** is inserted into a heart chamber of a patient P. Further processes can be performed in order to prep the patient for a diagnostic procedure, such as a process selected from the group consisting of: applying one or more surface electrodes **225** to the patient; preparing one or more alternate imaging devices, such as imaging device **80** described hereabove; delivery of one or more drugs or other agents to the patient, such as a heart medication or blood thinner; preparing ETC **210** for use; and combinations of two or more of these.

[0135] In STEP **520**, the system **10** is placed in a diagnostic mode. The diagnostic mode can be configured to produce one or more images or sets of information correlating to the anatomical shape and/or configuration of a heart chamber, and/or the electrical activity of a heart chamber, such as mapping information gathered prior to and/or during a cardiac ablation procedure. The diagnostic mode can comprise STEPS **530**, **540**, and **550**, performed repeatedly, simultaneously, or in a particular pattern, as described herein.

[0136] In STEP **530**, system **10** performs an analysis of biopotential data, determining dipole, surface charge and/or other voltage or charge based information correlating to the electrical activity of the heart, such as described in U.S. Pat. No. 8,417,313, entitled “Method and Device for Determining and Presenting Surface Charge and Dipole Densities on Cardiac Walls,” which is incorporated herein by reference. Electrodes **132** are electrically connected to ETC **210** of electronics module **200** via conduits **106** and cables **206**. ETC **210** can comprise one or more algorithms for determining dipole density and/or surface charge based on data recorded from electrodes **132**. ETC **210** can further comprise one or more filters (e.g. hardware or software filters), configured to pass (e.g. not significantly filter) biopotential signals, while filtering other signals, specifically ultrasound and/or localization signals present within the chamber of the heart or otherwise within Patient P. In some embodiments the processes of STEP **530** can be continuously performed during the completion and/or repetition of STEPS **540** and **550**, such as continuously while system **10** remains in a diagnostic mode.

[0137] In STEP **540**, a localization process is performed, such as a localization process described below with reference to FIG. 3.

[0138] In STEP **550**, an ultrasound measurement process is performed, such as an ultrasound measurement process described below in reference to FIG. 4.

[0139] In Step **560**, if system **10** remains in a diagnostic mode, STEPS **530**, **540**, and **550** are repeated. In some embodiments, such as when STEP **530** is continuously performed while system **10** remains in a diagnostic mode, STEPS **540** and **550** are repeated continuously while system **10** remains in a diagnostic mode. In some embodiments STEP **540** can be performed for longer, or multiple times for a single STEP **550**. In some embodiments STEP **550** can be performed for longer, or multiple times for a single STEP **540**.

[0140] System **10** can be placed in an alternate mode, such as a mode selected from the group consisting of: a hold mode, such as a mode when catheter **100** remains inserted in Patient P, however diagnostic procedures are not performed; an alert mode, such as a mode when system **10** has detected an error and diagnostic and/or other procedures are halted; a shutdown/completion mode, such as a mode when system **10** is deactivated, such as to be removed from Patient P at the end of a diagnostic or treatment procedure. In STEP **560**, when system **10** is determined to no longer be in a diagnostic mode, process **500** enters STEP **570**. In STEP **570**, all diagnostic procedures are stopped.

[0141] In some embodiments, system **10** can alternate between STEP **540** and STEP **550**, such as to gather localization information and ultrasound information to generate a model of the anatomy of the heart. Subsequently, STEP **530** and STEP **540** can be performed, alternately or simultaneously, such as to map the electrical activity of the heart, such that system **10** can register

the mapped electrical activity to the modeled anatomy gathered previously. In some embodiments, system **10** can again alternate between STEP **540** and STEP **550** to update the model of the anatomy.

[0142] Referring now to FIG. **3**, provided is an embodiment of a method of performing a localization process, consistent with the present inventive concepts. In some embodiments, process **600** of FIG. **3** is accomplished using system **10** of FIG. **1** described hereabove. In STEP **610**, system **10** begins a localization process. In some embodiments this process can be interleaved with an ultrasound measurement process as described below with reference to FIG. **4**.

[0143] In STEP **620**, signal generator **211** generates one or more localization signals, transmitted to patient P through one or more surface electrodes **225** via conduits **226**. Surface electrodes **225** can comprise one or more pairs of electrodes **225**, such as three pairs of electrodes **225**, configured to provide a three axis localization system. For example, in a three axis localization configuration, pairs of surface electrodes **225** can be placed on patient P; a first pair placed on the chest and back of patient P defining a first, X axis; a second pair placed laterally on the sides of patient P defining a second, Y axis; and a third pair placed on the neck or shoulder and thigh of patient P, defining a third, Z axis. Alternatively, a first pair of electrodes can be placed laterally on the sides of the patient defining a first axis, a second pair of electrodes can be placed on the upper chest and lower back of the patient defining a second axis, and a third pair of electrodes can be placed on the upper back and lower chest of the patient, defining a third axis. In some embodiments, signal generator **211** generates 3 or more signals of different frequencies, such as to drive three or more axes (e.g. each axis X, Y, and Z described hereabove), each at a unique frequency. The three or more axes can comprise two or more axes that are orthogonal to each other. Alternatively or additionally, signal generator **211** can generate 3 signals which differ in phase or other measurable characteristics, such that each signal (axis) can be determined via filtering to perform multi axis localization as describe herebelow. In some embodiments, each axis is powered individually (e.g. one at a time), and single axis localization can be interleaved between one or more desired axes. In the embodiment of process **600**, STEP **620** can be performed continuously, throughout process **600**, or throughout a diagnostic procedure (e.g. localization signals are continuously driven throughout the diagnostic procedure).

[0144] In STEP **630**, ETC **210** records data collected from one or more electrodes **132**, such as from each electrode **132** simultaneously or sequentially. In STEP **640**, the recorded data can be filtered one or more times, such as by one or more sequential filters and/or one or more parallel filters. In an embodiment, the recorded data can be initially filtered to isolate signals correlating to the localization signals generated by generator **211**, such as signals comprising a frequency between 1 and 100 kHz, such as between 10 and 100 kHz. The filtered data can subsequently be split and filtered by multiple (e.g. three) parallel filters, each configured to isolate a single frequency range, such as a frequency range associated with a single axis.

[0145] In STEP **650**, the three sets of individually filtered data can be analyzed, for example by a localization algorithm, such as to determine the location of each electrode **132**, in a three dimensional coordinate system relative to Patient P. In some embodiments, localization process **600** can comprise the use of more or fewer axes, such as two, three, or four axes. Additionally or alternatively, localization process **600** can comprise the use of concentric surface electrodes **225**. Localization process **600** can comprise multiple filters and/or multiple data paths within ETC **210**, such as multiple data paths corresponding to multiple axes, and multiple levels of data filtering.

[0146] In STEP **660**, if system **10** remains in a localization process, STEPS **620** through **650** are repeated. In some embodiments system **10** can remain in a localization process for a time period between 1 μ s and 1 s, such as between 50 μ s and 0.5 s, such as approximately 10 ms, for example when localization process **600** is interleaved with an ultrasound measurement process and each process is performed during similar or dissimilar amounts of time. In STEP **660**, when system **10** is determined to no longer be in a diagnostic mode, process **600** enters STEP **670**. In STEP **670**, the

localization **600** process is stopped.

[0147] Referring now to FIG. **4**, provided is a method of performing an ultrasound measurement process, consistent with the present inventive concepts. In some embodiments, process **700** of FIG. **4** is accomplished using system **10** of FIG. **1** described hereabove. In STEP **710**, system **10** begins an ultrasound measurement process. In some embodiments this process can be interleaved with a localization process as described in reference to FIG. **3** hereabove.

[0148] In STEP **720**, UTC **220** “activates” a first transducer **133** (which can be referred to as **133.sub.FIRST**), such as by closing one or more switches, electrically connecting the first transducer **133.sub.FIRST** to generator **221** and/or other electrical components of UTC **220**, such as is described in reference to FIG. **6** herebelow. In some embodiments, the one or more switches can comprise an opto-coupler, such as an opto-coupler with an activation time of approximately 0.01 μ s, or approximately 500 μ s. Generator **221** can be configured to generate a pulsed “drive signal”, configured to “ring”, vibrate, and/or otherwise cause transducer **133** to generate an ultrasonic pulse. The drive signal can comprise a signal with one or more frequencies between 1 MHz and 25 MHz, such as a drive signal with at least a frequency of approximately 10 MHz. The drive signal can further comprise a signal comprising a pulse width between 0.1 μ s and 10 μ s, such as a pulse width of approximately 1.0 μ s or 2.0 μ s.

[0149] In some embodiments, such as the paired electrode/transducer embodiment of FIG. **1**, the activation of a transducer **133** causes the deactivation of its paired electrode **132**. During the activation period of a transducer **133**, ETC **210** does not record electrical signals received by the paired electrode **132**, causing a temporary “blind spot”. As described herebelow, a non-sequential sequence of transducers **133** can be activated, such that the temporary “blind spot” in electrical recording is not extended by sequentially activating adjacent pairs **132/133**.

[0150] In STEP **730**, first transducer **133.sub.FIRST** remains activated, however is no longer being driven by generator **221**. Transducer **133** “rings down” (or is “rung down”), such as to allow all driven vibration of first transducer **133.sub.FIRST** to cease and any remnant vibrations within first transducer **133.sub.FIRST** to dissipate. In some embodiments, STEP **730** can comprise a duration of between 0.05 μ s and 1 μ s, such as a duration of approximately 0.1 μ s.

[0151] In STEP **740**, UTC **220** is configured to “listen”, such as by recording any ultrasonic vibrations sensed by first transducer **133.sub.FIRST**, and recording reflections of one or more ultrasonic pulses generated in STEP **720**. These reflections can correlate to reflections of ultrasound off of features or structures selected from the group consisting of: an inner wall of the cardiac chamber; an outer wall of the cardiac chamber; a feature of the cardiac chamber, such as a pulmonary vein or cardiac valve; a portion of a device inserted into the cardiac chamber, such as an ablation catheter and/or second mapping catheter also inserted into the cardiac chamber; and combinations of two or more of these. In some embodiments, STEP **740** can be configured to “listen” for reflections during a time period of between 1 μ s and 200 μ s, such as a time period of approximately 100 μ s. UTC **220**, or another component of electronics module **200**, can be configured to determine a distance measurement, such as a measured distance from first transducer **133.sub.FIRST** to the source of the first received reflection, such as a reflection from the inner wall of the cardiac chamber. The distance measurement can be determined using techniques commonly known to those skilled in the art, such as by determining the total “travel time” of the ultrasonic pulse, and using the speed of sound in blood and/or other tissue (as appropriate) to determine the total travel distance of the pulse.

[0152] In STEP **750** a subsequent transducer, **133.sub.NEXT** can be electronically prepared. Preparation can include “activating” transducer **133.sub.NEXT**, as described hereabove. STEP **750** can further comprise the deactivation of the previous transducer **133.sub.PREV**, for example transducer **133.sub.FIRST**. In some embodiments, activation of transducer **133.sub.NEXT** can comprise a process requiring a duration of between 0.01 μ s and 500 μ s, such as a duration of approximately 50 μ s. In these embodiments, the activation of transducer **133.sub.NEXT** can be

interleaved with a deactivation of the previous transducer **133.sub.PREV**, and/or with a portion of **STEP 740**, such that transducer **133.sub.NEXT** is being activated while transducer **133.sub.PREV** is listening and or being deactivated. In some embodiments, these processes can overlap for a time period of between 0.01 μ s and 500 μ s, such as a time period of approximately 100 μ s. In some embodiments, the duration from the start of an activation process of a transducer **133** to the end of a deactivation process can be between 1 μ s and 700 μ s, such as a duration of approximately 200 μ s. [0153] In **STEPS 760** through **780**, transducer **133.sub.NEXT** is rung, rung down, and listened to and recorded, as described in reference to **STEPS 720** through **740** hereabove.

[0154] In **STEP 790**, if all transducers **133** (or a predetermined subset thereof) have not been activated since the start of process **700**, **STEPS 750** through **790** are repeated. In some embodiments, a subset of transducers **133** are activated per process **700**, such as approximately half or approximately one third of the transducers **133**, such as when two or three cycles of process **700** are required to activate all transducers **133**, such as two or three cycles run sequentially or are interleaved with one or more other processes, such as process **600** of **FIG. 3**. In some embodiments, a complete cycle, such as a cycle in which all transducers **133** are activated, can comprise a duration of between 500 μ s and 10,000 μ s, such as a duration of approximately 5,000 μ s. In the embodiment of **FIG. 5** described herebelow, catheter **100** can comprise **48** transducers **133**. Each activation period can comprise a duration of approximately 200 μ s, and process **700** can comprise a duration of approximately 5 ms.

[0155] In **STEP 790**, if all transducers **133** (or a predetermined subset thereof) have been activated, process **700** continues to **STEP 795**. In **STEP 795**, if the measurement process is to be repeated, for example if a subsequent (similar or dissimilar) subset of transducers **133** is to be activated, **STEPS 720** through **790** are repeated. If the measurement process is completed, process **700** enters **STEP 799**. In **STEP 799**, the measurement process is stopped.

[0156] Referring now to **FIG. 5**, provided is a perspective view of an embodiment of a diagnostic catheter that includes expandable assembly **130**, consistent with the present inventive concepts. The expandable assembly **130** can be, in whole or in part, in accordance with the description of U.S. patent application Ser. No. 14/762,944, entitled "Expandable Catheter Assembly with Flexible Printed Circuit Board (PCB) Electrical Pathways", filed Jul. 23, 2015, which is incorporated herein by reference. In the embodiment of **FIG. 5**, the expandable assembly **130** includes a plurality of splines **131** configured as shown (i.e., six splines, radially separated by **60°**, each spline comprising eight electrode transducer pairs **132/133**). In this embodiment, transducers **133** are coupled to splines **131** using a housing **138**. In other embodiments, multiple transducers **133** can be coupled to splines **131** (e.g. between two and twelve splines **131**) in different manners.

[0157] In this embodiment, an array of transducers **133** and electrodes **132** are substantially equally distributed across splines **131**, as shown in the expanded state of expandable assembly **130**.

Proximal ends (nearest shaft **105**) of splines **131** are attached to a distal portion of shaft **105**, such as at a location in and/or within shaft **105**, or between shaft **105** and an inner, translatable (i.e. advanceable and retractable) shaft, control rod **107**. Control rod **107** can comprise one or more conduits and/or passageways, such as lumen **108** as shown. Lumen **108** can be configured to allow for catheter **100** to be inserted over a guidewire, such as when lumen **108** is sized to slidably receive a guidewire, and lumen **108** continues to a proximal portion of catheter **100**, such as when lumen **108** exits handle **110** of catheter **100**. Additionally or alternatively, lumen **108** can be sized to slidably receive one or more devices such as a device selected from the group consisting of: an ablation catheter; a mapping catheter; a cryo ablation catheter; a tip ablation catheter; a diagnostic catheter; and combinations of two or more of these. In some embodiments, lumen **108** can be configured to allow for the delivery of one or more drugs or other agents during a diagnostic or other procedure.

[0158] In some embodiments, electrodes **132** can be positioned on the inside of splines **131**. Alternatively or additionally, electrodes **132** can comprise some electrodes positioned on the inside

of splines **131** and some electrodes positioned on the outside of spline **131**. Alternatively or additionally, electrodes **132** can be double sided electrodes, with opposing surfaces facing both inward and outward of the basket, or electrodes **132** can comprise ring-shaped electrodes, surrounding each spline **131** respectively.

[0159] As shown, distal ends of splines **131** are connected to the distal end of control rod **107**. Control rod **107** can be advanced and retracted to compact and expand, respectively, expandable assembly **130**. Control rod **107** can be advanced and retracted via a control on a proximal handle, such as control **111** on handle **110** of FIG. **1**. In some embodiments, control rod **107** can be retracted from a position correlating to the natural expanded position of expandable assembly **130** (as shown by example in FIG. **5**), such as to deform expandable assembly **130**, such as to invert a distal portion of splines **131**, resulting in at least the distal most transducers **133** aligning in a forward facing direction, as shown in FIG. **5A**. In this configuration, the forward facing transducers **133** can be used as an array of transducers to perform B mode scans, or other ultrasound scanning methods known in the art.

[0160] As described herein, expandable assembly **130** of FIG. **5**, including forty eight electrode/transducer pairs **132/133**, can be used to perform biopotential measurements, localization measurements, and/or ultrasound distance measurements. During an ultrasound measurement process, such as process **700** of FIG. **4** described hereabove, transducers **133** of expandable assembly **130** of FIG. **5** can be sequenced as described herebelow in reference to FIG. **6**.

[0161] Referring now to FIG. **6**, a representation of an activation sequence of an array of **48** ultrasound transducers disposed on six splines (**8** per spline) is illustrated, consistent with the present inventive concepts. FIG. **6** is a particular representation of an activation sequence, representing a specific number of transducers, substantially equally spaced across a specific number of splines on an expandable assembly, such as expandable assembly **130** of FIG. **5** hereabove. Alternatively, expandable assembly **130** can have different numbers of transducers and/or splines, and a similar or dissimilar non-sequential sequence of transducer activation can be performed.

[0162] In the embodiment of FIG. **6**, transducers 1-8 represent a most distal (1) transducer through a most proximal transducer (8), across each of six splines. Each activation period depicted by a solid box represents a period of activation and as described herein, a deactivation or blanking of a paired electrode. The pattern shown represents a pattern avoiding the sequential activation of two neighboring transducers, such as a pattern avoiding the sequential activation of two transducers within two or three “neighboring spaces” of each other. Neighboring spaces can be considered spaces on a single spline; across splines, such as transducer 1 of spline 1 and transducer 1 of spline 2; and/or diagonally across splines, such as transducer 1 of spline 1 and transducer 2 of spline 2. The pattern shown also represents a pattern avoiding sequential activation of two transducers from a single spline.

[0163] FIG. **7** provides an embodiment of a block diagram of a user interface (UI) system **230** that can be used with a diagnostic catheter as described herein, for example, in accordance with the present inventive concepts.

[0164] The UI system **230** includes a display area **240**, which can include one or more windows, screens, and/or monitors on which information can be rendered/shown, e.g., as 2D or 3D displays. The windows in the display area **240** need not be arranged nor relatively sized as shown in FIG. **7**. And not all windows shown in display area **240** must be included. The depiction in FIG. **7** represents an illustrative embodiment, but a UI system in accordance with the inventive concept is not limited to the particular embodiment shown.

[0165] A 3D display window **242** can be included to show graphical elements in a three-dimensional (3D) space, such as a heart or heart chamber. The images and information rendered in the 3D display window **242** can change based on the user task being performed, e.g., based on the task being done in a main application window **250**. The 3D display window **242** can also exist

within the main application window **250**, in some embodiments. The 3D display window **242** can be user interactive, and can change in response to the user interaction therewith.

[0166] A two-dimensional (2D) display window **244** can be included to show graphical elements in a two-dimensional space. The images and information rendered in the 2D window **244** can change based on the user task being performed, e.g., based on the task being done in the main application window **250**. The 2D display window **244** can also exist within the main application window **250**, in some embodiments. The 2D display window **244** can be user interactive, and change in response to the user interaction therewith.

[0167] The main application window **250** can include the primary workflow interface to create 3D maps. An acquisition window **252** provides tools, e.g. user interface tools, necessary to view and record biopotential signals, localization signals, and/or ultrasound signals. One tool of the acquisition window **252** allows ultrasound and localization data to be combined to reconstruct a chamber anatomy (i.e. build a digital model of a surface that represents the chamber anatomy). This representation of the anatomy can be displayed in a surface building window **254**. Additionally, previously reconstructed chamber anatomies (e.g. of the patient and/or a surrogate) can be loaded from one or more data repositories, such as files, databases, or memory and displayed in the surface building window **254** to be used with live data. Configuration settings are available from this window **254** to properly register/orient a chamber reconstruction to the live data.

[0168] A waveform processing window **256** can be provided and used to allow recorded data to be reviewed, filtered, and/or analyzed. The user can use these tools to identify a time segment of data to be mapped. Segments can be from **1** sample in length to the full recorded data length. Segment selection can also take the form of passing data directly, time sample by time sample, to the mapping algorithm such that maps can be made “on the fly” (e.g. in real-time or near real-time, or pseudo real-time, “real-time” herein), without manual segment selection. The waveforms being processed can be shown in the 2D display window **244**, e.g., in the form of an electrogram (EGM) or electrocardiogram (ECG or EKG). The 3D display window **242** can show any or all of the following: the voltage signals on the basket electrodes rendered onto a three-dimensional surface of the size and shape of the basket, a colored topographic surface showing the electrode signals (color and “Z-height” of the topography corresponding to voltage amplitude), with electrodes oriented in relative neighbor relationship, and/or the spatial position of the basket in relation to the reconstructed surface to show the basket position within the chamber of interest.

[0169] A mapping window **258** can be provided and used to allow configuration and execution of the mapping algorithms, including selection of a surface source model. The resulting 3D maps can be rendered in the 3D display window **242** with corresponding waveforms shown in the 2D display window **244**. A time cursor or window can be included to provide a time index between display windows. The time cursor or window can be configured to slide or move across the waveforms in the 2D window in synch with a dynamically changing display rendered in the 3D window.

[0170] A system configuration and diagnostic window **246** can be provided and used to show live signals from the catheters (e.g., processed through electronics module **200**)-biopotential, localization, and/or ultrasound, as examples. This window **246** can be used for verification of operation of such systems or subsystems.

[0171] A surface editing window **248** can be provided and used to allow the user to edit and process the reconstructed anatomy. Tools provided can include but are not limited to: selection (individual vertices/polygons, rectangular, elliptical, free-form shape, automatic isolated component selection and/or sharp feature selection), trimming (through-cut, front-surface cut), smoothing, re-meshing, hole-filling, sub-division, and surface deformation, such as push-pull, tools. These tools can include shape identification, component identification, isolation, extraction, appending and/or merging tools. These tools can be user interactive surface editing tools. These tools can be configured to operate manually, semi-automatically and/or automatically.

[0172] A user input module **260** can include human interface devices, such as mouse, keyboard,

touchscreen, digital pen, or other devices that can be used to provide user input to and/or control of the system and its renderings.

[0173] FIGS. **8A-8C** provide different views relating to the output of the user interface system, in accordance with aspects of the present inventive concepts.

[0174] Referring to FIG. **8A**, a point cloud (PointCloud) data structure is shown, which can be rendered in the 3D display window **242**. According to this embodiment, the 3D coordinate space is divided into spherical sectors with quadrilateral cross-sections, except the poles which are N-sided. The cross-section of each bin at the same radius from the origin is configured to be similar in area. Surface point coordinates fall into one and only one bin, so do not overlap. Bin size, e.g., subtended azimuth or elevation angle, can be configurable (e.g. on instantiation). To change bin-size (and thus mesh size) and/or displacement of the surface relative to the center of the spherical bins, all surface points in an existing PointCloud can be placed into a second data structure with the desired parameters in one bulk operation.

[0175] A surface representative of the surface points in the data structure is displayed by merging all representative points or surface of each bin. In one embodiment, the representative vertices can be drawn with the interconnecting mesh between bins to form the surface. As points are added to the data structure, bins will be updated and the representative surface is updated correspondingly. Bins with no points within them can be hidden from display.

[0176] Referring to FIG. **8B**, a PointCloud bin is shown and described with reference to a 3D rendering of a heart. All data points falling in each bin are analyzed to determine a representative point (vertex) or surface (surface patch) for the bin. In one embodiment, the centroid of all points in the bin is used as a representative vertex. Data within each bin can be assessed for quality, and vertices or polygons of the representative surface can be colored to indicate quality of the data. In one embodiment, the dispersion or radial distance variance in the data can indicate the detection of a cardiac valve, vein, or other radially-oriented anatomical structure.

[0177] Referring to FIG. **8C**, a subset of neighboring bins are shown, and their relationships illustrated, where each bin is represented by a block. A non-manifold interconnecting mesh is calculated between neighboring bins. The orientation relationship of bins is static to avoid time-consuming recalculation of the non-manifold interconnecting mesh between neighbors.

[0178] FIG. **9** provides an embodiment of a functional block diagram of a cardiac information processing system **900**, in accordance with the present inventive concepts.

[0179] Using the described system from FIG. **9**, a user can choose what to calculate and/or what to display, e.g., the user can display Dipole Density (DDM), Charge Density (CDM), or Voltage (V-V). This information is calculated based on information represented in the top three boxes **902**, **904**, **906**, e.g., the position of the electrodes **902**, the shape and location of the chamber (surface) **904**, and the potentials recorded at the electrodes **906**. The system can also be configured to support and enable changes back and forth between the different display modes, and with post processing tools, can change how that information is displayed.

[0180] The processing includes selecting a forward model **908**. Based thereon, one of the following three operations can be performed: Dipole Density Mapping (DDM) **910**, Charge Density Mapping (CDM) **912**, and/or Voltage to Voltage Mapping (V-V) **914**. In Dipole Density Mapping (DDM), electrical fields that could be measured by electrodes inside and/or outside of the heart chamber are generated from a distribution of dipole sources, having a magnitude and direction, on the surface of the heart chamber, organized and arranged as Dipole Densities (DD). In Charge Density Mapping (CDM), electrical fields that could be measured by electrodes inside or outside of the heart chamber are generated from a distribution of scalar charge sources, having a magnitude only, on the surface of the heart chamber, organized and arranged as Charge Densities (CD). And in Voltage to Voltage Mapping (V-V), no source assumption is made, and the voltages measured on electrodes inside or outside of the heart chamber are propagated from the voltages on the heart chamber surface (e.g. using Laplace's equation and/or other methods known to those skilled in

electromagnetic field theory).

[0181] With the chamber surface and electrodes' positions registered with the surface as the inputs, the transform matrix, which encodes relationships between the DD/CD/Voltages on the heart chamber to the measured voltages on electrodes, is the output of the forward calculation.

[0182] An Inverse Calculation **916** is performed, with the potentials acquired from the mapping catheter and the transform matrix (the output from the forward calculation) as the inputs, the DD/CD/Voltages on the surface can be obtained by solving a linear system using a regularization method, for example the Tikhonov regularization method.

[0183] DD/CD/Voltages on the surface **920** are outputs from the inverse calculation **916**. The surface voltages can be forwardly computed from the derived surface DD/CD for DDM/CDM, and surface voltages from V-V can be used to derive the surface DD/CD using the transform matrix specified by the heart chamber surface.

[0184] In some embodiments, cardiac information processing system **900** comprises post-process tool **930**. Using the same, DD/CD/Voltages can be post-processed to produce a Coulombian map (an adaptation of the discrete Laplacian, or spatial second derivative of the DDM, CDM and/or Voltage maps), IsoChrono map (activation timings), Magnitude map (peak to peak magnitude or negative peak magnitude), Persistence map (active and resting status), and/or Propagation map (the wavefront), as examples.

[0185] The 3D Display **242** can be used to display the outputs from the post-processing tools **930**. That is, for example, surface DD/CD/Voltages, as well as post-processing maps, can be rendered by selecting options on the display panel of UI system **230**. The 3D maps can be rotated to different viewing angles and a color map can be adjusted by a user, as examples.

[0186] While the foregoing has described what are considered to be the best mode and/or other preferred embodiments, it is understood that various modifications can be made therein and that the invention or inventions may be implemented in various forms and embodiments, and that they may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim that which is literally described and all equivalents thereto, including all modifications and variations that fall within the scope of each claim.

Claims

1. (canceled)

2. A body cavity imaging system, comprising: a catheter configured for delivery to a body cavity defined by surrounding tissue: a plurality of ultrasound transducers coupled to a distal end of the catheter, wherein the plurality of ultrasound transducers are coupled to a 3D array: an electronics module configured to selectively turn on/off each ultrasound transducer according to a predetermined activation sequence and to process signals received from each ultrasound transducer to produce a 3D display of the surrounding tissue.

3. The system of claim 2, wherein the body cavity is a heart chamber and the surrounding tissue is one or more walls of the heart chamber.

4. The system of claim 2, wherein the 3D display of the surrounding tissue is presented on a user interface system having a display screen and user control mechanism enabling graphical manipulation of the 3D display of the surrounding tissue.

5. The system of claim 2, wherein the 3D array is a basket array, spiral array, a balloon, radially deployable arms, and/or other expandable and compactible structures.

6. The system of claim 2, wherein the ultrasound transducers are disposed on a plurality of splines of the 3D array.

7. The system of claim 6, wherein the 3D array includes at least three splines.

8. The system of claim 6, wherein at least two ultrasound transducers are disposed on each spline.

9. The system of claim 6, further comprising a plurality of biopotential electrodes coupled to the

distal end of the catheter, wherein the biopotential electrodes are also disposed on the plurality of splines of the 3D array.

10. The system of claim 9, wherein at least some of the biopotential electrodes and at least some of the ultrasound transducers are disposed on the same splines.

11. The system of claim 9, wherein a biopotential electrode and an ultrasound transducer are disposed together to form an electrode/transducer pair, and the system includes a plurality of electrode/transducer pairs.

12. The system of claim 11, wherein one or more splines comprise a plurality of electrode/transducer pairs.

13. The system of claim 11, wherein a plurality of splines comprise at least one electrode/transducer pair.

14. The system of claim 11, wherein each spline comprises a flexible PCB, and each electrode/transducer pair is electrically coupled to the flexible PCB.

15. The system of claim 11, wherein each electrode/transducer pair shares a common communication path.

16. The system of claim 11, wherein all electrode/transducer pairs on a spline share a common communication path.

17. The system of claim 11, wherein all electrode/transducer pairs on a spline share a common ground.

18. The system of claim 2, further configured to correlate cardiac or other electrical activity to one or more images generated using an imaging device.

19. The system of claim 18, wherein the imaging device comprise an imaging device selected from the group consisting of: a fluoroscope; an MRI; a CT Scanner; an ultrasound imaging device; and combinations of two or more of these.

20. The system of claim 2, wherein the activation sequence comprises a pattern of turning on/off the plurality of ultrasound transducers that avoids a sequential activation of two neighboring ultrasound transducers.

21. The system of claim 20, wherein the activation sequence avoids the sequential activation of two transducers within two or three neighboring spaces of each other.
